

# Network, Coverage & Technology

What powers BharatConnect — and how to explain it to customers

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This document explains BharatConnect's network technology in two registers: a plain-language version agents can relay to customers, and a technical reference for accuracy. When talking to a customer, always lead with the plain-language version.

## 1. Network Generations We Run

| Technology | Status on BharatConnect                              | What the customer gets                                 |
|------------|------------------------------------------------------|--------------------------------------------------------|
| 2G / 3G    | Being phased out; voice fallback only in a few areas | Calls/SMS where 4G unavailable                         |
| 4G LTE     | Nationwide — the backbone of the network             | Reliable data + Volte HD calling everywhere we operate |
| 5G         | Live in 180+ cities (NSA + expanding SA)             | Much faster data, lower lag in covered areas           |
| 6G         | NOT available — does not commercially exist yet      | Nothing today; future/experimental only                |

**Important: there is no 6G**

6G is still a research/standardisation topic worldwide and is **not commercially available from any operator**, including BharatConnect. Expected availability is around the 2030s.

If a customer asks for or claims to have '6G', politely clarify that BharatConnect offers **4G nationwide and 5G in 180+ cities**, and that 6G is a future technology not yet launched anywhere.

## 2. BharatConnect 5G

### 2.1 Where it's available

- Live in 180+ cities, led by Pune, Mumbai, Chennai, Bengaluru, Hyderabad, Delhi NCR, and expanding into Tier-2 cities each quarter.
- Coverage within a city is not uniform — it is strongest in dense urban zones and along major corridors.
- Customers can check live 5G coverage for their area on the website coverage map.

### 2.2 How a customer gets 5G

- A **5G-capable handset** with the latest software update.
- An **active 5G-eligible plan** (most current unlimited/daily plans include 5G at no extra cost).
- Being **physically in a 5G coverage area**; the phone shows '5G' in the status bar when connected.
- 5G is enabled automatically — there is no separate '5G recharge'.

### 2.3 Technical reference

- **Architecture:** Launched on 5G NSA (Non-Standalone, anchored on the 4G core); migration to 5G SA (Standalone) underway for lower latency and network slicing.

- **Spectrum:** Mid-band 3.3–3.6 GHz (capacity + coverage balance) and select mmWave (26 GHz) hotspots in dense areas; low-band 700 MHz for wide/indoor coverage.
- **Typical speeds:** 200–700 Mbps in good mid-band conditions; higher in mmWave hotspots.

### 3. BharatConnect 4G LTE

- The nationwide backbone — present everywhere BharatConnect operates, including most rural and Tier-3 areas.
- Carries **VoLTE** (Voice over LTE) for HD voice calls; this is why calls and data work simultaneously.
- **Bands:** 900 MHz, 1800 MHz, 2100 MHz, 2300 MHz, and 700 MHz for deep indoor/rural coverage.
- **Typical speeds:** 15–50 Mbps depending on location and congestion.

### 4. BharatConnect Fiber (Wired Broadband)

Separate from the mobile network: Fibre-to-the-Home (FTTH) using GPON technology, delivering symmetric-ish high speeds to homes and offices in covered cities.

| Fiber plan | Speed    | Typical use                                     |
|------------|----------|-------------------------------------------------|
| Fiber-100  | 100 Mbps | Everyday browsing, HD streaming, work-from-home |
| Fiber-300  | 300 Mbps | 4K streaming, multiple devices, gaming          |
| Fiber-500  | 500 Mbps | Heavy households, content creators              |
| Fiber-1G   | 1 Gbps   | Power users, small offices                      |

### 5. Network Troubleshooting (Agent Quick Guide)

| Symptom                     | Likely cause                                       | What to advise                                                                                      |
|-----------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| No network / no signal      | Coverage gap, outage, SIM seating, airplane mode   | Toggle airplane mode; reseal SIM; check known outage for area; if none, raise fault ticket + docket |
| Slow data despite full bars | FUP throttling, congestion, 4G vs 5G area          | Check if daily/FUP quota used (KB-POL-002 §3); suggest add-on; confirm 5G area if expecting 5G      |
| Phone shows 4G, not 5G      | Outside 5G area, non-5G handset, plan, or software | Confirm device + plan + location are all 5G-ready; check coverage map on website                    |
| Calls drop frequently       | VoLTE off, weak indoor coverage, handset issue     | Ensure VoLTE enabled in phone settings; try another location; raise ticket if persistent            |
| Fiber down                  | Local fibre cut, router/ONT power, area outage     | Check router/ONT lights & power; check area outage; raise fiber fault ticket with docket            |

### 6. How to Talk About the Network

- Lead with what the customer gets, not the spectrum band. ("You have fast 5G in your area" rather than "we run 3.5 GHz mid-band NSA".)
- Be honest about coverage — if 5G isn't in their area yet, say so and point to the coverage map.
- Never overpromise speeds; quote typical ranges, not peak lab figures.
- Reinforce: 5G needs a 5G phone + 5G plan + 5G area, and there is **no extra charge** and **no separate 5G recharge**.